Cu incubation study

Raw recordings to reviewed events and publication figures

Follow one page at a time. Orange numbers on each screenshot match the numbered actions beside it. Numbers restart on the next page.

01

Prepare

Identify each ABF and apply the exact recording recipe.

02

Review

Check detections; optionally edit individual events or whole clusters.

03

Analyze & style

Recompute, make the complete figure set, style it and save.

Choose your route. Keep the automatic event decisions to reproduce the saved reference. For a curated analysis, use a separate workspace copy and rerun the affected downstream stages.

The Cu reference uses model-free Schmitt detection. No trained model is required.

Launcher and shared curation views are refreshed for build 40.15.59. Unchanged reference illustrations retain their capture dates. Examples show saved or explicitly labeled fixture data; this documentation update did not rerun the scientific studies.

Find the next step

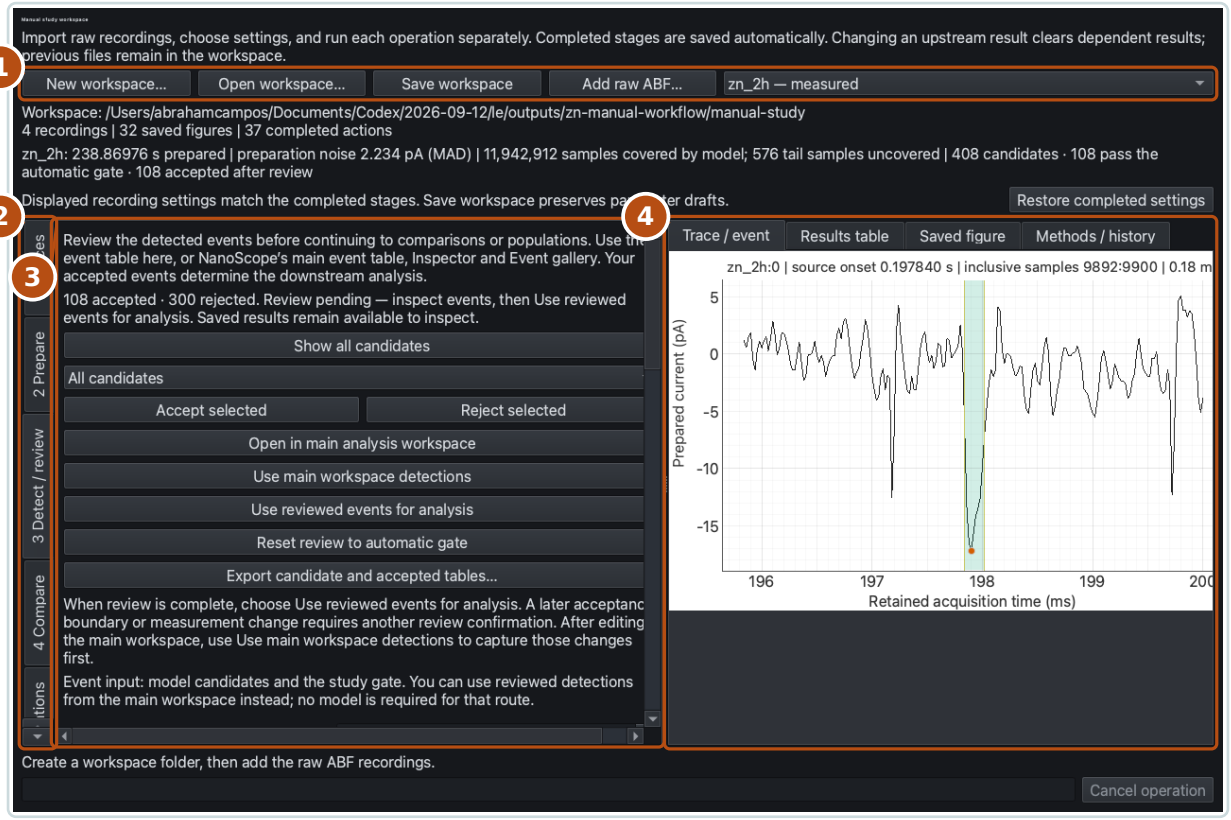
Use the page IDs in the text to find the corresponding walkthrough or recipe.

cu01	Create or reopen an incubation study	4
cu02	Add recordings to your new study	5
cu03	Set each recording’s identity	6
cu04	Reconstruct the primary 30-hour current	7
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First, see the workspace overview on page 3. Recipes and checkpoints follow the walkthrough: exact file names, scientific settings, counts and the figure inventory. Use the PDF bookmarks to jump directly to any walkthrough or recipe.

Find your way around the workspace

Study controls and result views. Use the following close-ups for the current launcher and shared event-curation steps.



Reference capture, 21 Sep 2026. NanoScope reference controls and saved results.

1 Choose the recording

The top toolbar creates, opens and saves the workspace; the selector chooses the active recording.

2 Choose the stage

The six tabs run from Sources to Figures. If a tab is hidden, use the small arrow at the bottom of the tab strip.

3 Set and run

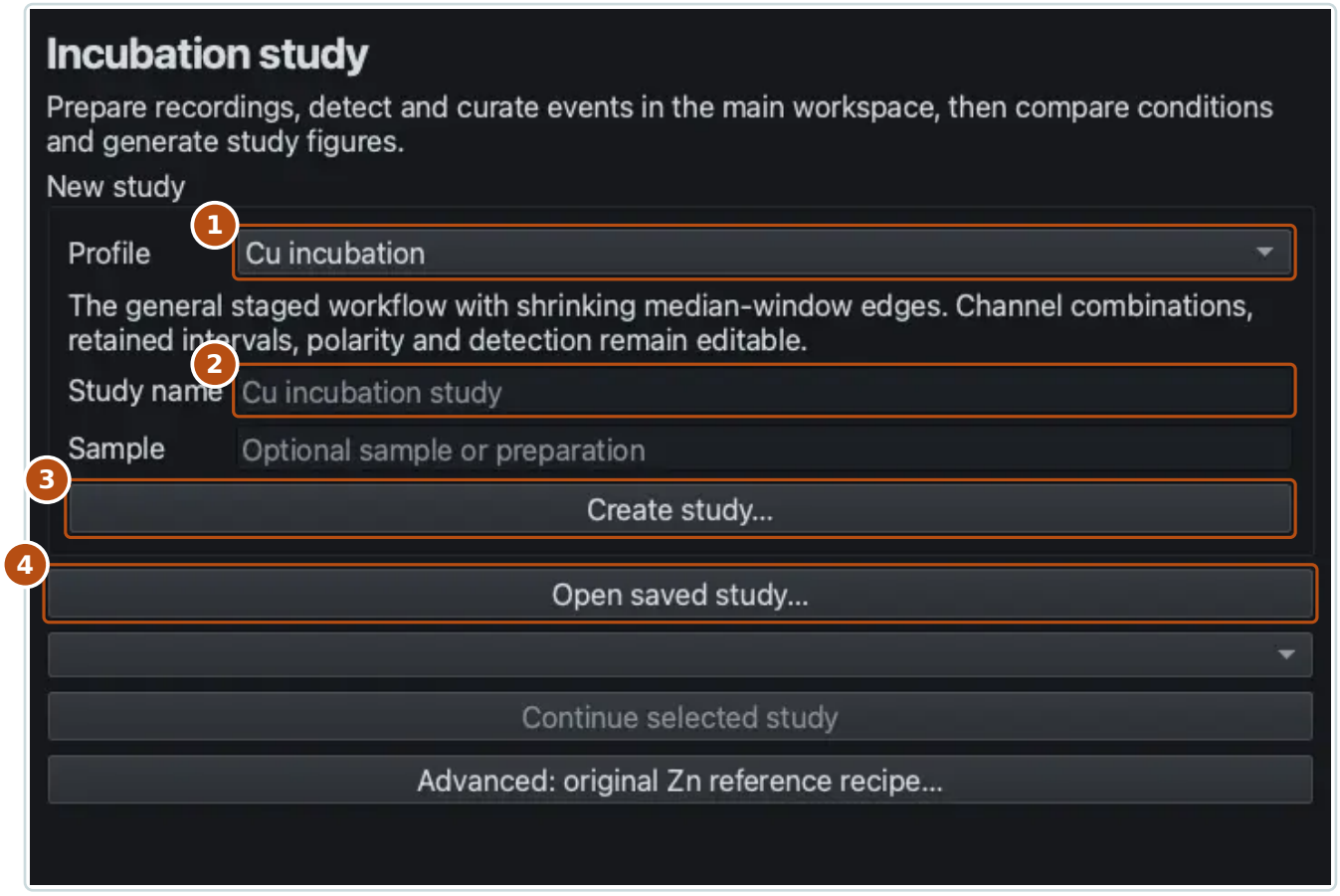
Enter values and click the stage button here. Scroll inside this panel to reach the lower controls.

4 Inspect the result

Switch among the trace, tables, figures and methods here. These tabs display the saved results.

Create or reopen an incubation study

Unified incubation workflow | NanoScope 40.15.59



Current unified study launcher · cu

DO THIS IN NANOSCOPE

1. **Choose the profile**
   Open Analysis > Incubation study... in the main app. Choose Cu incubation. Use General incubation for another sample or condition.

2. **Identify this study**
   Enter a useful study name and the sample, such as Cu-insulin. These labels do not infer or validate experimental settings.

3. **Create a new study**
   Click Create study... and choose a new empty folder. This creates the workspace; do not create it a second time in the study window.

4. **Or reopen a saved study**
   Click Open saved study... and choose the folder containing workspace.json. NanoScope restores its saved method and settings automatically.

Keep in mind

The profile supplies starting defaults. Apply every recording-specific recipe on the following pages. Saved workspaces retain their own numerical settings.

You should see

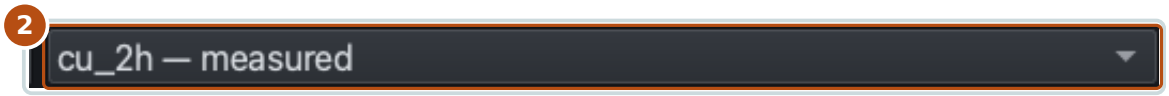
A named study opens with its staged controls. Continue selected study returns to an already open workspace.

Add recordings to your new study

Setup · a new folder for a fresh reproduction



Reference capture, 21 Sep 2026. Workspace toolbar



Reference capture, 21 Sep 2026. Workspace toolbar

DO THIS IN NANOSCOPE

1. **Add the source recordings**
   Click Add raw ABF… and select the five exact files in Recipe A. Preserve their original names.

2. **Select the recording to edit**
   Use the recording selector to work on one file at a time. The six stage tabs act on this selection.

3. **Save the starting workspace**
   Click Save workspace. Completed operations will also save automatically.

Keep in mind

Create study… on the launcher has already created this folder. Add the exact reference ABFs listed in the input appendix. Choose a separate saved-workspace copy for alternative curation.

You should see

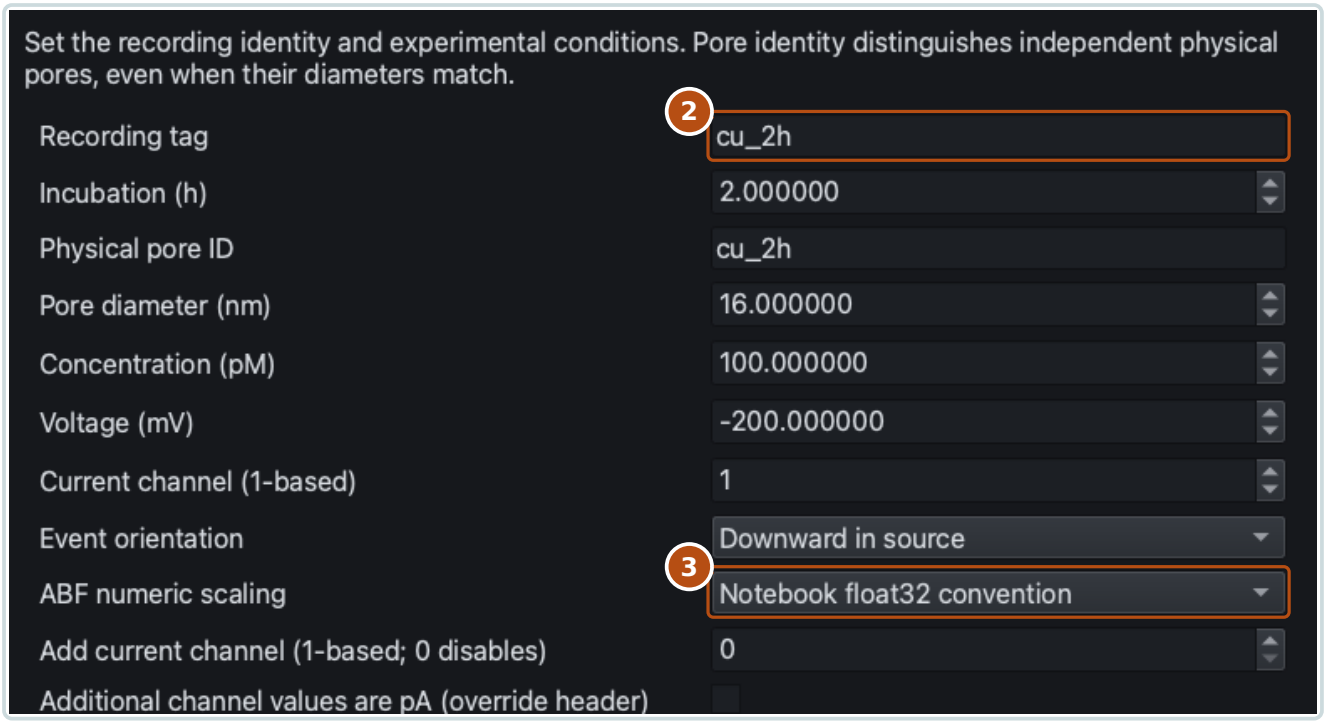
Five recordings appear in the selector. You have your own workspace folder, separate from the delivered reference.

Set each recording’s identity

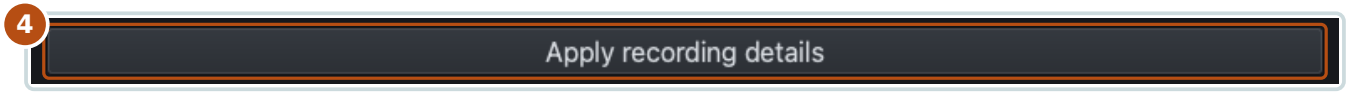
1 Sources · repeat for all five recordings



Reference capture, 21 Sep 2026. Workspace toolbar



Reference capture, 21 Sep 2026. Sources · top of panel



Reference capture, 21 Sep 2026. Scroll down · apply recording details

DO THIS IN NANOSCOPE

1. **Choose the recording**
   Select the source file in the top recording selector.

2. **Enter its experimental details**
   In 1 Sources, enter the tag, incubation, physical pore ID and pore diameter from Recipe B. Use concentration 100 pM and voltage -200 mV.

3. **Set the source conventions**
   Use current channel 1, Downward in source, Notebook float32 convention and Shrinking (Cu convention). Recipe B lists all fields.

4. **Commit this recording**
   Click Apply recording details before selecting another recording. Complete the exceptions on the next three pages.

Keep in mind

All files are sampled at 50,000 Hz. Source interval times start at the first sample of the supplied file; a “10–15 min” filename or acquisition header does not offset these fields.

You should see

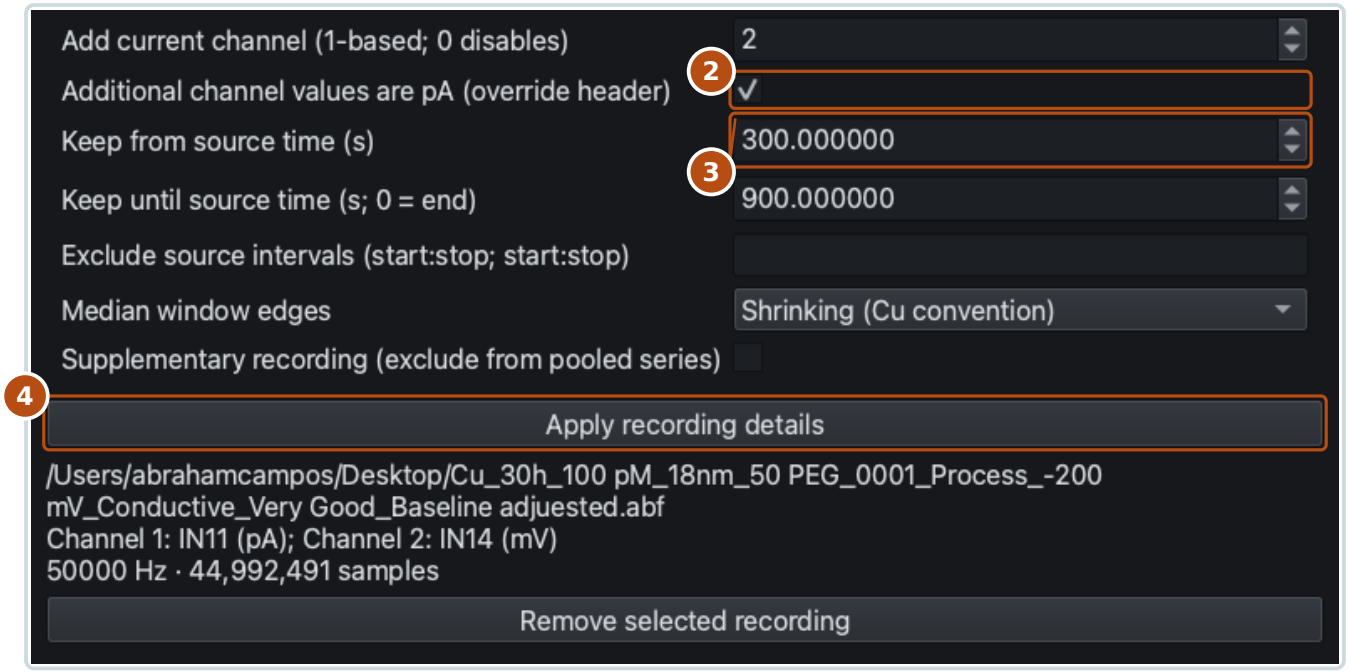
The saved recording has the correct tag and conditions. Matching pore diameters still have separate physical pore IDs.

Reconstruct the primary 30-hour current

1 Sources · cu_30h only



Reference capture, 21 Sep 2026. Workspace toolbar



Reference capture, 21 Sep 2026. Controls · scroll down

DO THIS IN NANOSCOPE

1. **Select the primary file**
   Choose cu_30h: the file ending Baseline adjuested.abf directly on the reference Desktop. It is not the 10–15-minute excerpt.

2. **Add the removed baseline**
   Set Add current channel to 2 and turn on Additional channel values are pA (override header).

3. **Keep the intended interval**
   Set Keep from source time to 300 and Keep until source time to 900. Leave exclusions blank and Supplementary recording off.

4. **Apply the settings**
   Click Apply recording details.

Keep in mind

In this processed export only, channel 1 is baseline-subtracted current and channel 2 holds the removed baseline. Its mV header is misleading; use its numeric values as pA to reconstruct the current. This is not a general voltage-to-current conversion.

You should see

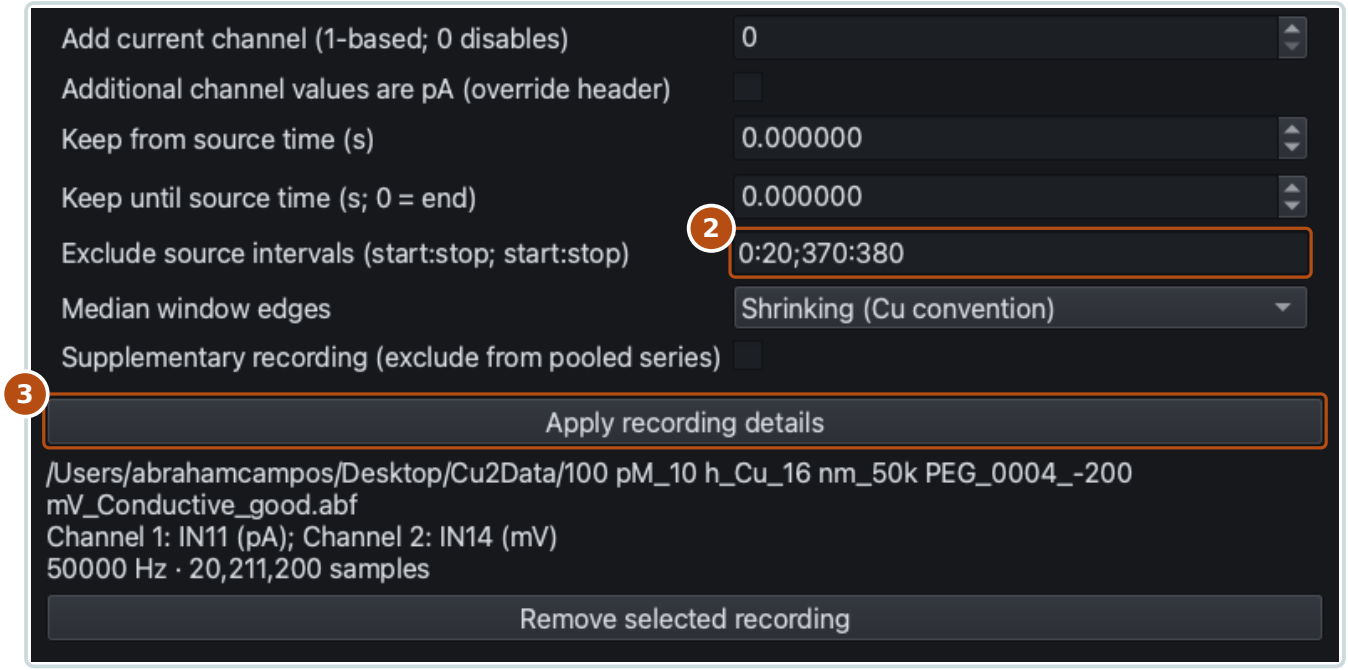
Preparation will retain 29,992,491 samples and 599.84982 s. The final source sample is at 899.84980 s.

Remove the two 10-hour intervals

1 Sources · cu_10h only



Reference capture, 21 Sep 2026. Workspace toolbar



Reference capture, 21 Sep 2026. Controls · scroll down

DO THIS IN NANOSCOPE

1. **Select cu_10h**
   Choose the 10-hour recording in the top selector.

2. **Enter both exclusions**
   Type 0:20;370:380 in Exclude source intervals. Keep the source range at 0 to 0, meaning the full file before exclusions.

3. **Apply the recording details**
   Leave added channel at 0 and Supplementary recording off, then click Apply recording details.

Keep in mind

Excluded intervals remove both endpoints. The baseline is calculated on the concatenated retained samples; detection and event merging cannot cross a removed source gap.

You should see

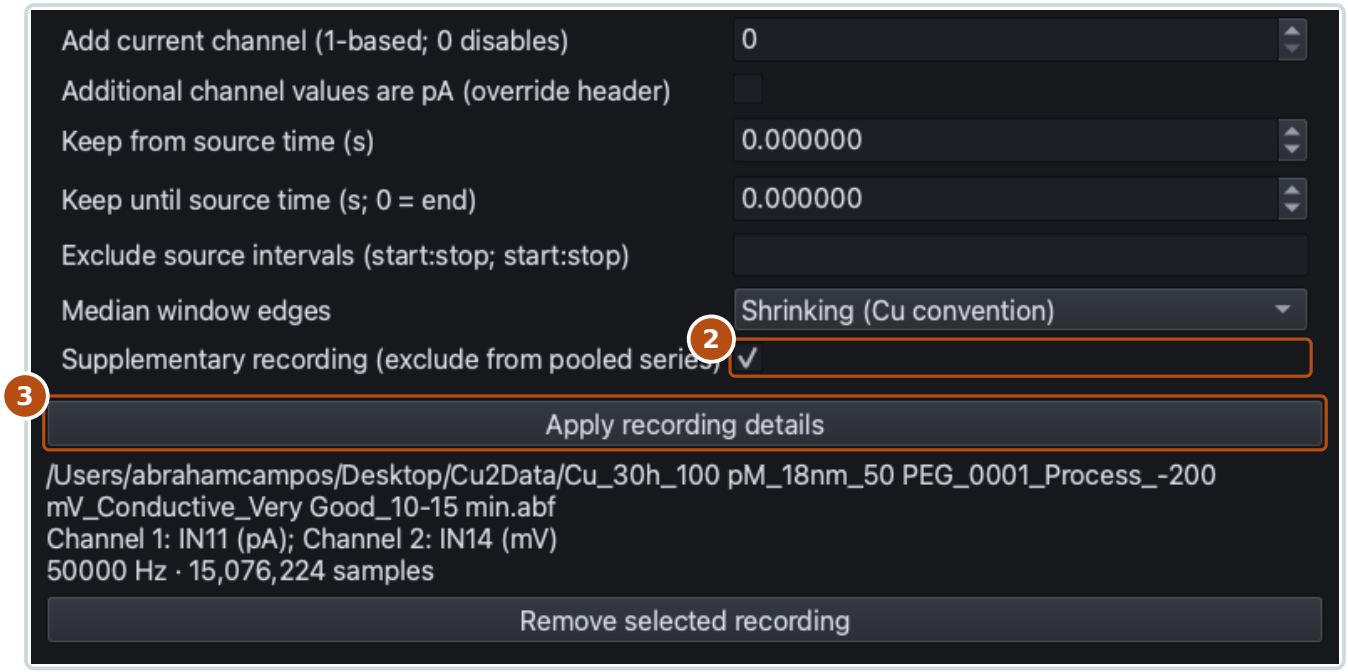
Preparation will retain 18,711,198 samples and 374.22396 s.

Keep the old excerpt supplementary

1 Sources · cu_30h_excerpt only



Reference capture, 21 Sep 2026. Workspace toolbar



Reference capture, 21 Sep 2026. Controls · scroll down

DO THIS IN NANOSCOPE

1. **Select the excerpt**
   Choose cu_30h_excerpt: the Cu2Data file ending 10-15 min.abf.

2. **Mark it as supplementary**
   Turn on Supplementary recording (exclude from pooled series).

3. **Apply the details**
   Use 30 h, pore ID cu_30h_excerpt, diameter 18 nm, full range 0 to 0, added channel 0 and no exclusions. Click Apply recording details.

Keep in mind

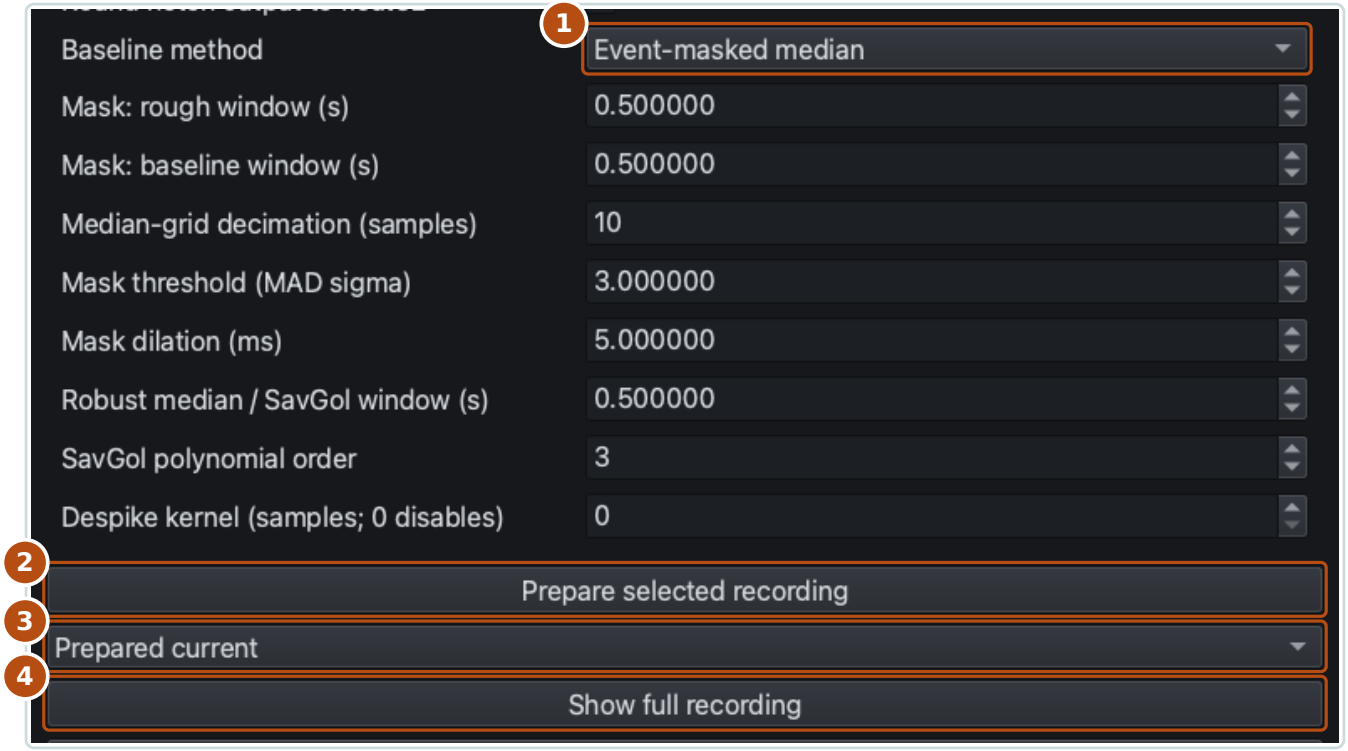
The excerpt is a separate recording-state comparison. Its raw open-pore level differs from the reconstructed primary 30-hour source; do not substitute one for the other.

You should see

All five recordings can be inspected, but the incubation comparison and pooled feature set use four primary recordings.

Prepare and inspect the current

2 Prepare · repeat for all five recordings



Reference capture, 21 Sep 2026. Prepare · baseline settings and action buttons

DO THIS IN NANOSCOPE

1. **Enter the preparation recipe**
   Open 2 Prepare. Select Event-masked median; set both mask windows to 0.5 s, decimation 10, threshold 3 and dilation 5 ms. Apply all Recipe C values.

2. **Run preparation**
   Click Prepare selected recording and wait for completion. Notch and despiking remain off.

3. **Inspect the baseline**
   Choose Raw current, then Filtered current and baseline in the preview selector. Check that the baseline follows drift, not the events.

4. **Check the prepared trace**
   Choose Prepared current. Use Show full recording to restore the overview, then repeat for each recording.

Keep in mind

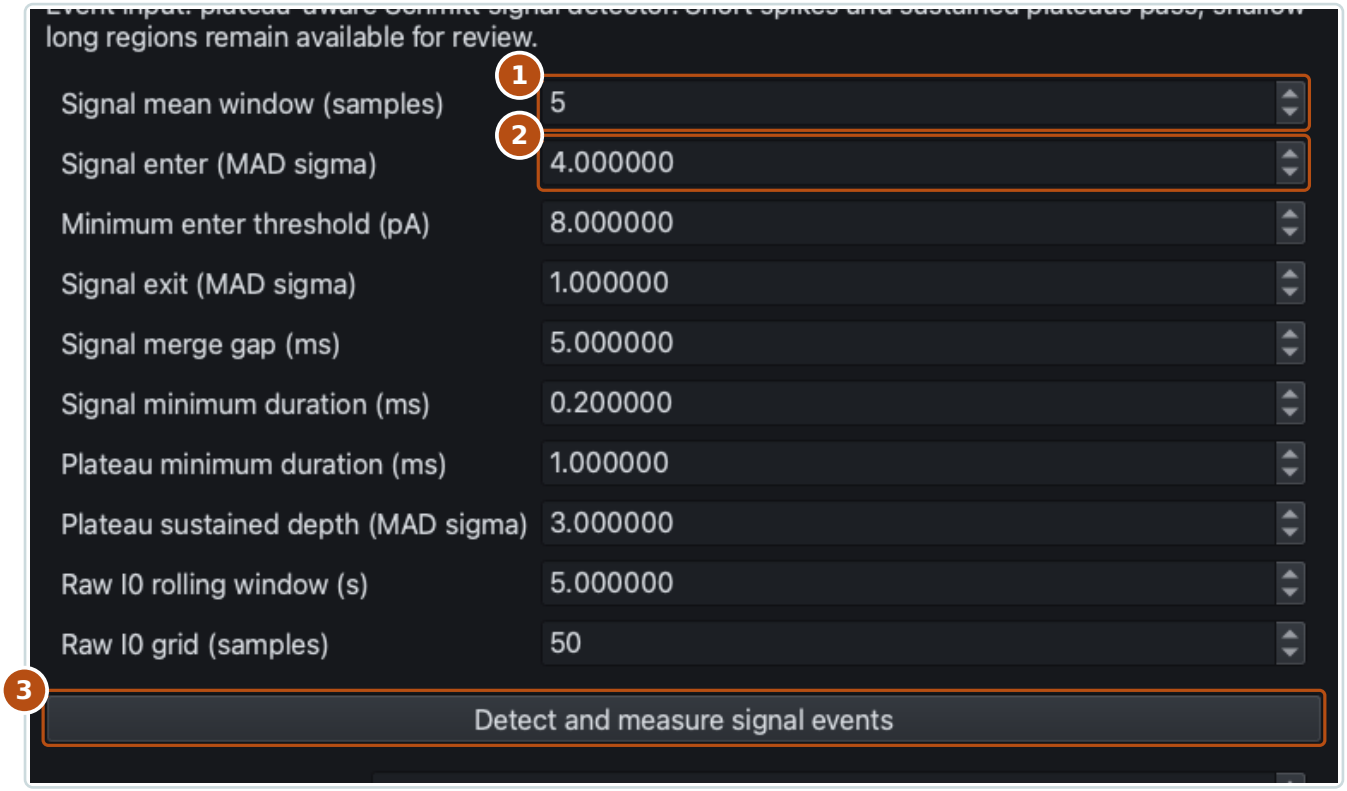
Cu requires a 0.5 s baseline window and 5 ms dilation; these differ from the general workspace defaults. Export prepared CSV and metadata… is optional; keep both files together.

You should see

Retained durations match Checkpoint 1. The baseline is plausible and each selected recording has a completed preparation.

Detect and measure Cu signal events

3 Detect / review · scroll down to signal controls



Reference capture, 21 Sep 2026. Controls · scroll down



Reference capture, 21 Sep 2026. Workspace toolbar

DO THIS IN NANOSCOPE

1. **Find the signal detector**
   In 3 Detect / review, scroll below the review buttons to Signal mean window (samples).

2. **Enter the detector recipe**
   Enter all ten Signal controls from Recipe D. Keep the mean window 5, entry 4 MAD sigma with an 8 pA minimum, and exit 1 MAD sigma.

3. **Run detection and measurement**
   Click Detect and measure signal events and wait. This one operation performs both stages for Cu.

4. **Repeat for each source**
   Use the top recording selector and repeat for the remaining four prepared recordings.

Keep in mind

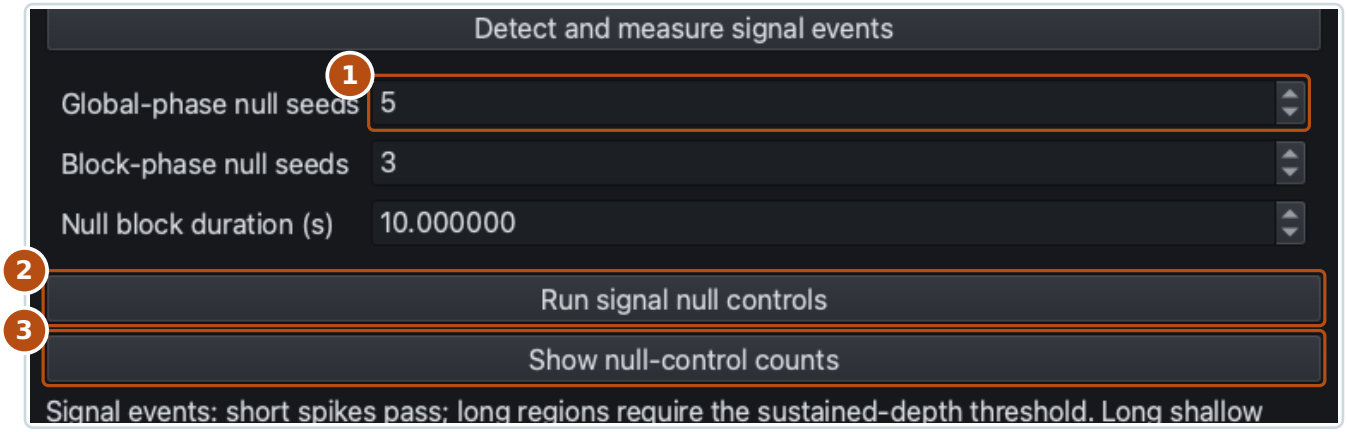
This is the Schmitt route; leave model inference, decoding and model-candidate measurement unused. Short spikes pass; regions ≥1 ms need sustained depth ≥3 MAD sigma.

You should see

Before manual changes, candidate / accepted counts are 60 / 34, 111 / 95, 161 / 58, 951 / 417 and 729 / 274 in source order.

Run the null-control diagnostics

3 Detect / review · repeat for all five recordings



Reference capture, 21 Sep 2026. Controls · scroll down



Reference capture, 21 Sep 2026. Workspace toolbar

DO THIS IN NANOSCOPE

1. **Set the null controls**
   Under the signal detector, set Global-phase null seeds 5, Block-phase null seeds 3 and Null block duration 10 s.

2. **Run the controls**
   Click Run signal null controls and wait for completion.

3. **Read the saved counts**
   Click Show null-control counts and inspect the Results table.

4. **Repeat for all five recordings**
   Switch the active recording and run the same controls.

Keep in mind

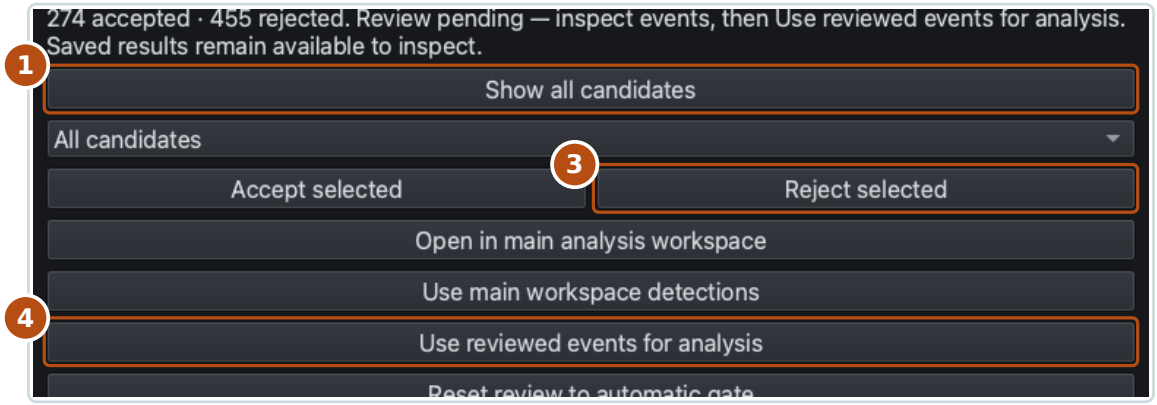
The surrogate thresholds are fixed from the real trace. These plots are diagnostics, not a measured false-discovery rate.

You should see

Each recording has nine surrogate rows: five global-phase, three block-phase and one sign-flip control.

Inspect and approve individual events

3 Detect / review · your selection drives later analysis



Reference capture, 21 Sep 2026. Controls · scroll down

2

	event_id	accepted	passes_gate	kind	onset_s	dwell_ms	lepth_sust_p	epth_peak_p	sta
1	cu_30h_ex...	True	True	plateau	1.1782	1.26	14.648438	134.8877	5891
2	cu_30h_ex...	False	False	shallow_lo...	2.18068	6.54	4.8828125	9.765625	1090
3	cu_30h_ex...	True	True	spike	3.99908	0.32	4.5776367	19.836426	1999
4	cu_30h_ex...	True	True	spike	4.17192	0.92	4.5776367	10.375977	2085
5	cu_30h_ex...	False	False	shallow_lo...	4.77078	1.54	4.9704208	10.153168	2385
6	cu_30h_ex...	True	True	spike	5.40284	0.68	4.119873	11.59668	2701
7	cu_30h_ex...	True	True	spike	5.67908	0.28	4.4250488	12.207031	2839

Reference capture, 21 Sep 2026. Saved reference result

DO THIS IN NANOSCOPE

1. **List the candidates**
   Click Show all candidates. Use the filter to view All candidates, Accepted only or Rejected only.

2. **Inspect a selected event**
   Click a row in Results table, then view Trace / event. Check its waveform, marked bounds and measurement.

3. **Make an event decision**
   Select one or several rows, then choose Accept selected or Reject selected. For reference reproduction keep the automatic decisions.

4. **Approve this recording**
   Click Use reviewed events for analysis. Repeat for all four primary recordings and the supplementary excerpt.

Keep in mind

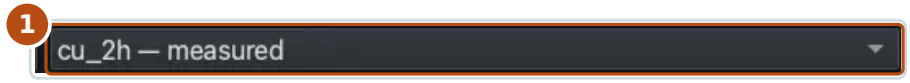
Accept/reject acts on selected rows, not every visible row. Reset review to automatic gate restores the initial selection and clears confirmation. Confirmation approves the selection; it does not claim that you inspected every event individually.

You should see

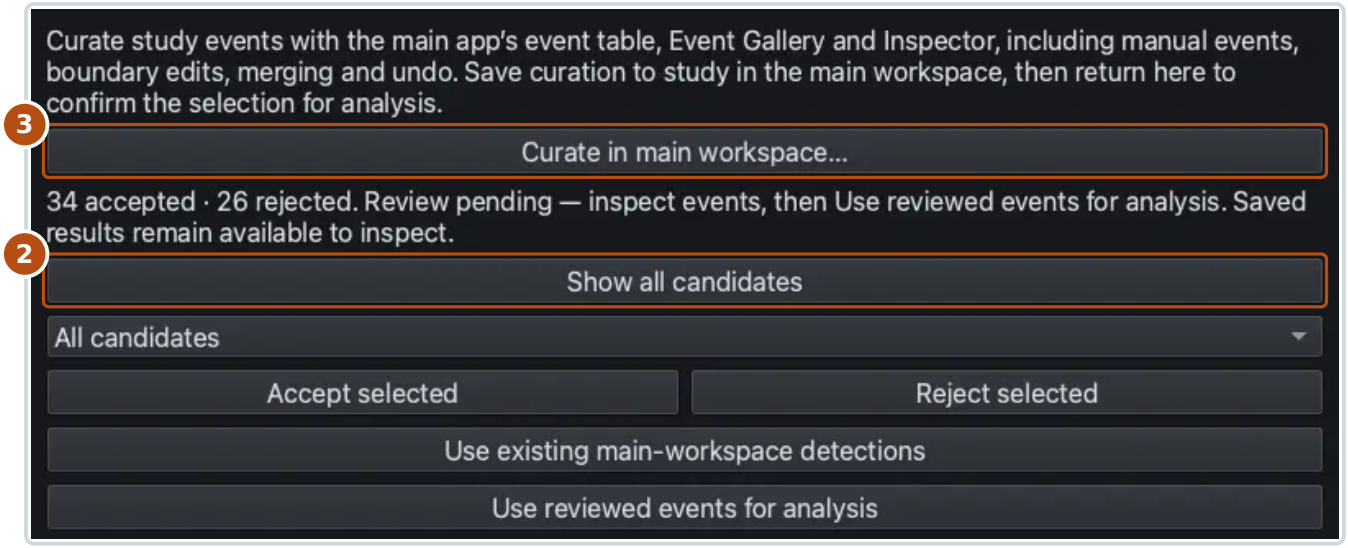
Review status is confirmed for every included recording. The unchanged reference has 604 accepted primary events; the excerpt has 274.

Curate with the main event tools

Unified incubation workflow | NanoScope 40.15.59



Current recording selector, cropped from the study toolbar.



Current build 40.15.59. CU review · Curate in main workspace; return and confirm selection

DO THIS IN NANOSCOPE

1. **Select the prepared recording**
   Use the recording selector to choose the source you want to review. Check its identity and finish preparation first.

2. **Inspect the study selection**
   Show the candidate table to check decisions before opening the main workspace. The main Gallery and Inspector provide detailed editing.

3. **Open the main workspace**
   Click Curate in main workspace.... The main app opens the matching prepared trace and available study events.

Keep in mind

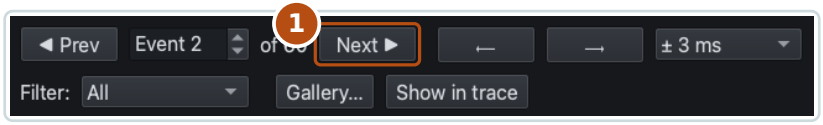
For a separate curated analysis, first save and copy the complete reference workspace. The prepared review trace is not the original open-pore current. Do not rerun detection if you only intend to review existing events.

You should see

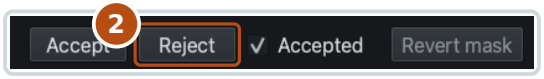
The main workspace carries an attached study bar with Save curation to study and Return to study.

Use the main Curate tools for closer inspection

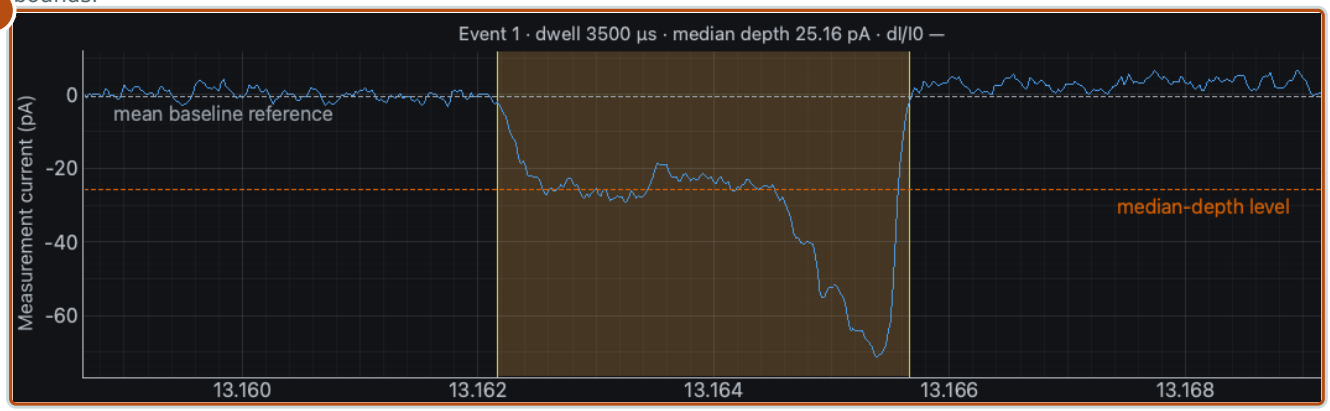
Optional · Event Inspector and boundary editing



Reference capture, 21 Sep 2026. Choose the event and context; use Prev / Next to inspect each detection.



Reference capture, 21 Sep 2026. Accept or reject the displayed event. Revert mask becomes available after editing its bounds.



Reference capture, 21 Sep 2026. A real cu_2h event. The shaded interval is the event mask; native measurements use the prepared trace.

DO THIS IN NANOSCOPE

1 Move between events
Use ◀ Prev and Next ▶, or the arrow keys. Check the event identity and the current filter.

2 Accept or reject this event
Click Accept or Reject. A/R shortcuts do the same; they do not advance. The Accepted checkbox shows the decision.

3 Correct a boundary if justified
Drag the event mask edge in Event Inspector. Wait for the edit and measurement to finish. Revert mask restores stored original bounds when available.

Keep in mind

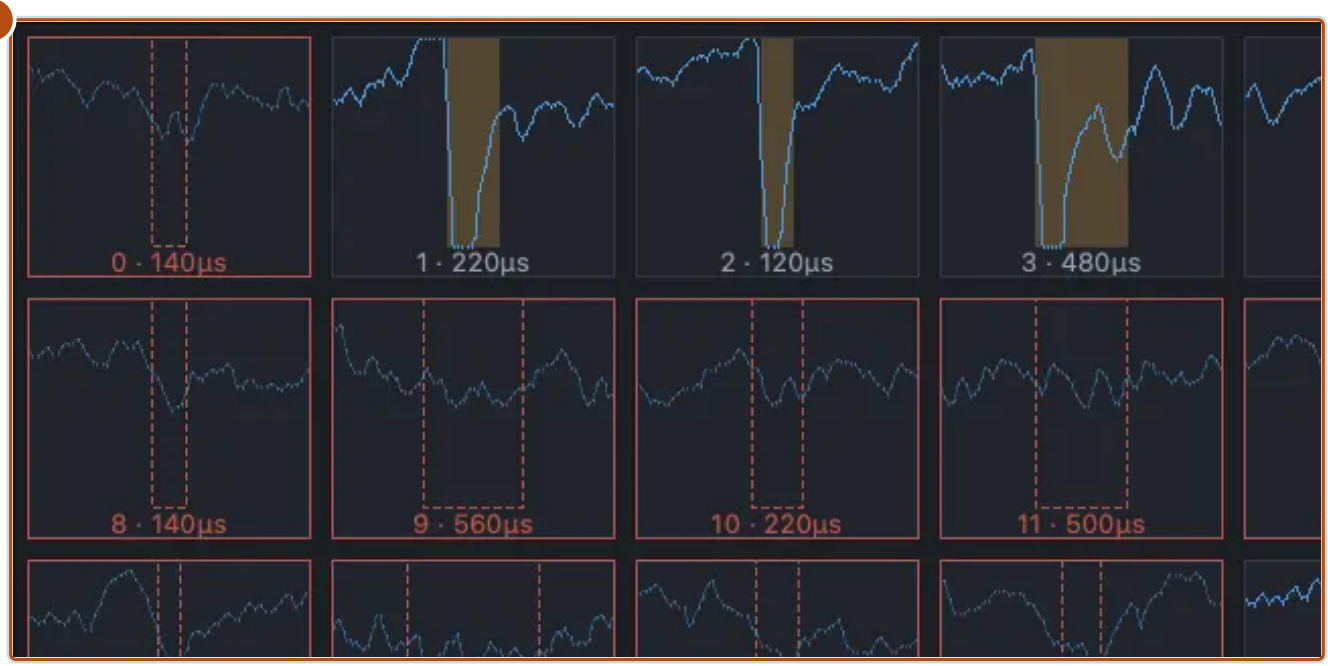
Open Event Inspector from the main Analysis menu after Curate in main workspace.... The study preview itself does not edit boundaries. Reconciled tables retain their source-editing restrictions.

You should see

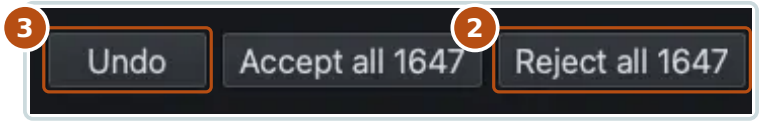
The main event table contains the intended acceptance and bounds. Next use Save curation to study and Return to study.

Review waveforms in the Event Gallery

Unified incubation workflow | NanoScope 40.15.59



Current Zn Gallery: cropped example tiles; remaining tiles are outside this crop.



Current Gallery bulk controls and Undo; read the full filter count.

DO THIS IN NANOSCOPE

1. **Compare the event tiles**
   From the main Analysis menu choose Event gallery…. Compare waveforms and open an event in the Inspector for precise boundary edits.

2. **Check the scope before a bulk decision**
   The all-events acceptance controls act on the current filter across pages. Read the displayed count. Use Accept selected or Reject selected for picked tiles instead.

3. **Undo an unintended edit**
   The Gallery Undo button uses the same undo history as the main workspace. Check the resulting event decisions after undoing.

Keep in mind

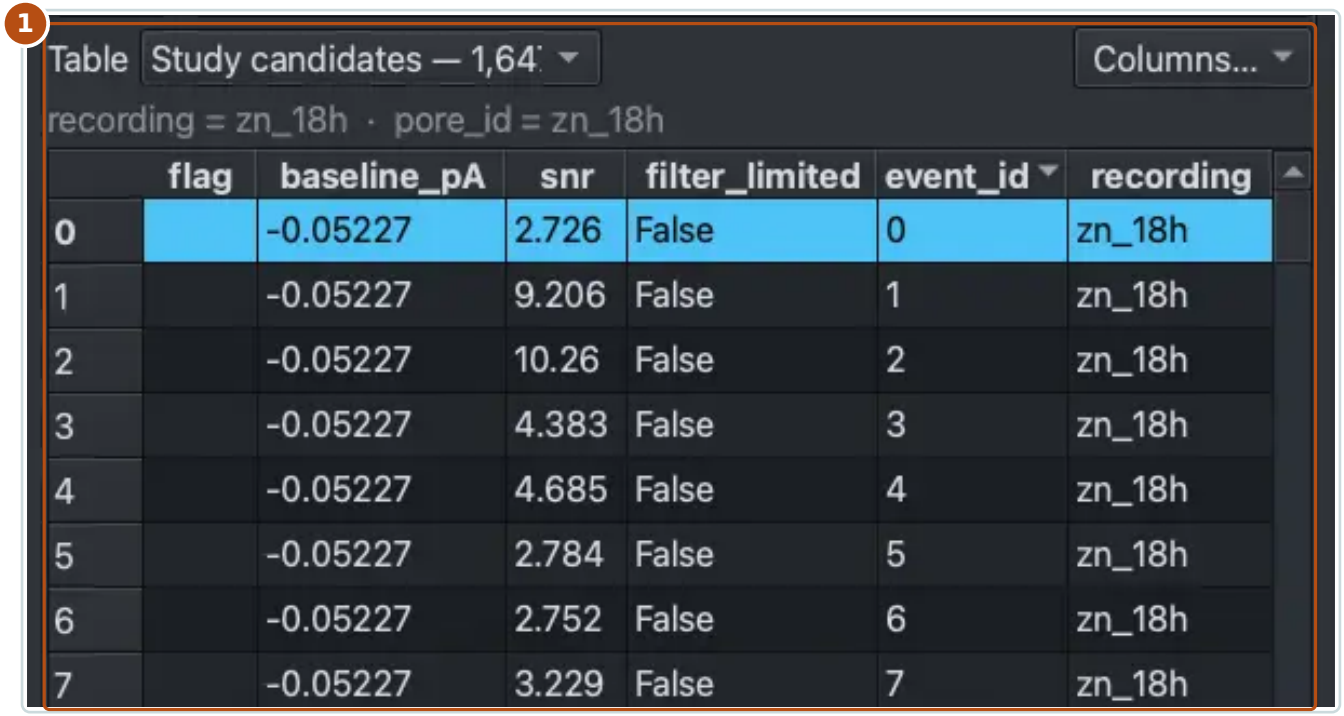
The screenshot is a cropped current-build view of saved Zn reference events. These controls are shared by all profiles. Reference reproduction retains its original acceptance decisions.

You should see

After Gallery edits, return to the main workspace and use Save curation to study before returning and confirming.

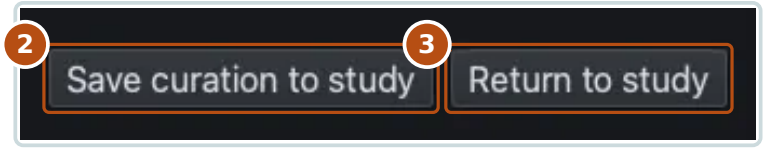
Save event edits back to the study

Unified incubation workflow | NanoScope 40.15.59



	flag	baseline_pA	snr	filter_limited	event_id	recording
0		-0.05227	2.726	False	0	zn_18h
1		-0.05227	9.206	False	1	zn_18h
2		-0.05227	10.26	False	2	zn_18h
3		-0.05227	4.383	False	3	zn_18h
4		-0.05227	4.685	False	4	zn_18h
5		-0.05227	2.784	False	5	zn_18h
6		-0.05227	2.752	False	6	zn_18h
7		-0.05227	3.229	False	7	zn_18h

Current event table, cropped to its identity and first rows.



Current study bar: save the curation before returning.

DO THIS IN NANOSCOPE

1. **Check the active events**
   Verify that the event table belongs to the recording you opened from the study. Review acceptance and edited bounds.

2. **Commit this curation**
   Click Save curation to study. Wait for successful completion; the study recalculates its measurements from its own raw and prepared signals.

3. **Return to the study**
   Click Return to study. Inspect the imported decisions and confirm the selection before recomputing affected analyses.

Keep in mind

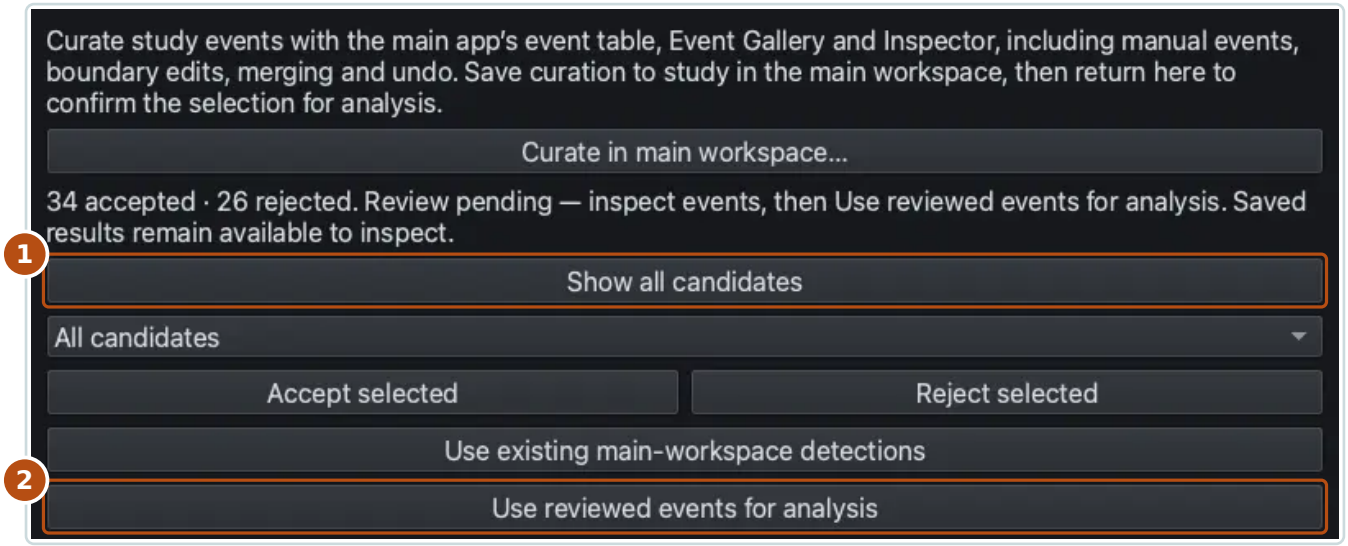
Save Project retains the main analysis session; it does not replace Save curation to study. Later event edits require another save. A failed or stale save is not a successful transfer.

You should see

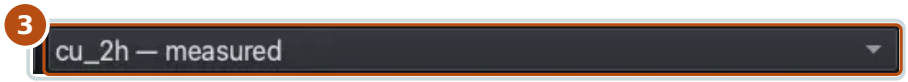
Acceptance and bounds are committed to the attached study recording. Dependent results must be recomputed after selection changes.

Confirm the reviewed selection

Unified incubation workflow | NanoScope 40.15.59



Current build 40.15.59. CU review · Curate in main workspace; return and confirm selection



Current recording selector, cropped from the study toolbar.

DO THIS IN NANOSCOPE

1. **Inspect the saved events**
   Click Show all candidates. Check several edited events, their bounds, acceptance and recording identity.

2. **Approve this selection**
   Click Use reviewed events for analysis. Repeat for each included recording whose current selection is pending.

3. **Repeat for the other recordings**
   Select the next recording and complete its preparation, review and confirmation before pooled analysis.

Keep in mind

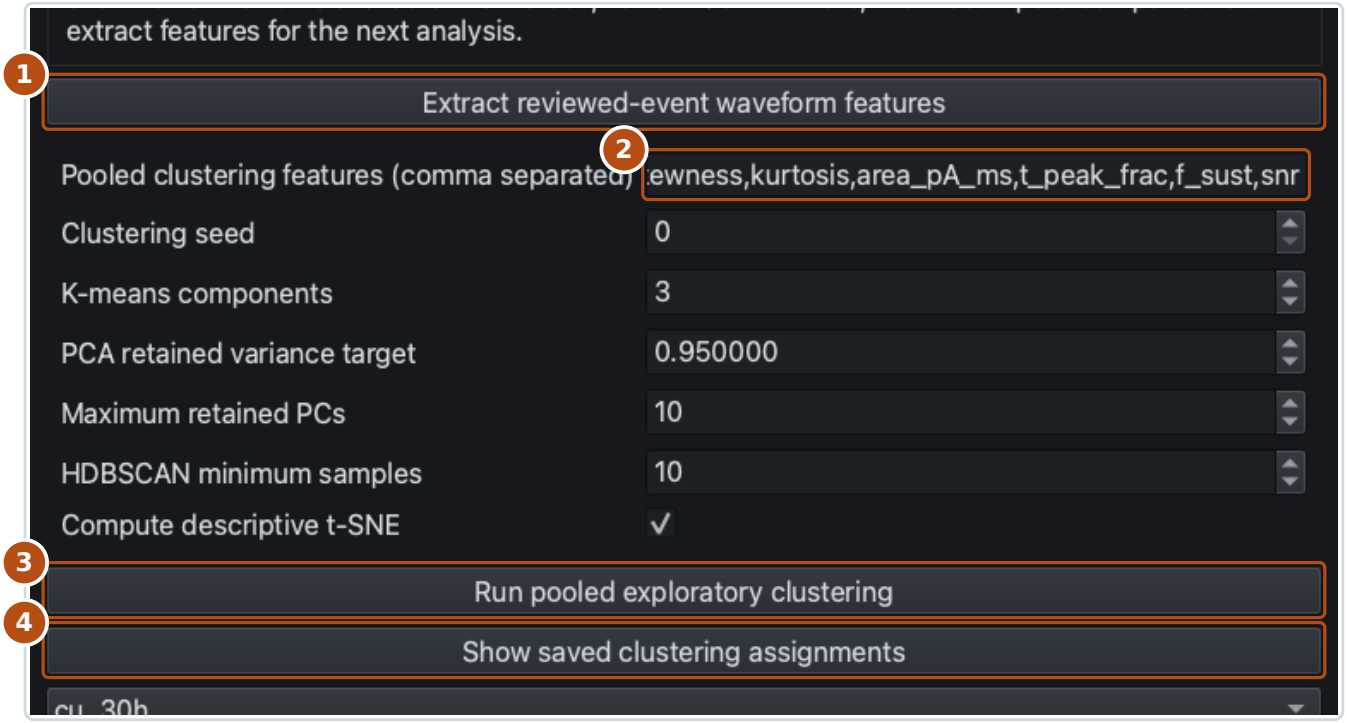
Confirmation approves the selection; it does not assert that every event was individually inspected. Automatic passes_gate remains distinct from accepted decisions. Reset review to automatic gate intentionally discards the current selection.

You should see

Recompute comparisons, features and fits from the final selection, then regenerate affected figures.

Extract features and form exploratory groups

5 Features / populations · final if no later selection changes



Reference capture, 21 Sep 2026. Controls · scroll down

DO THIS IN NANOSCOPE

1. **Extract the approved waveforms**
   Open 5 Features / populations and click Extract reviewed-event waveform features. Wait for completion.

2. **Set the pooled recipe**
   Paste Recipe F's exact 12-feature list in Pooled clustering features. Set seed 0, K-means 3, PCA target 0.95, maximum PCs 10, HDBSCAN minimum samples 10 and t-SNE on.

3. **Run the grouping**
   Click Run pooled exploratory clustering and wait.

4. **Inspect the committed assignments**
   Click Show saved clustering assignments; Show waveform feature table displays the feature rows.

Keep in mind

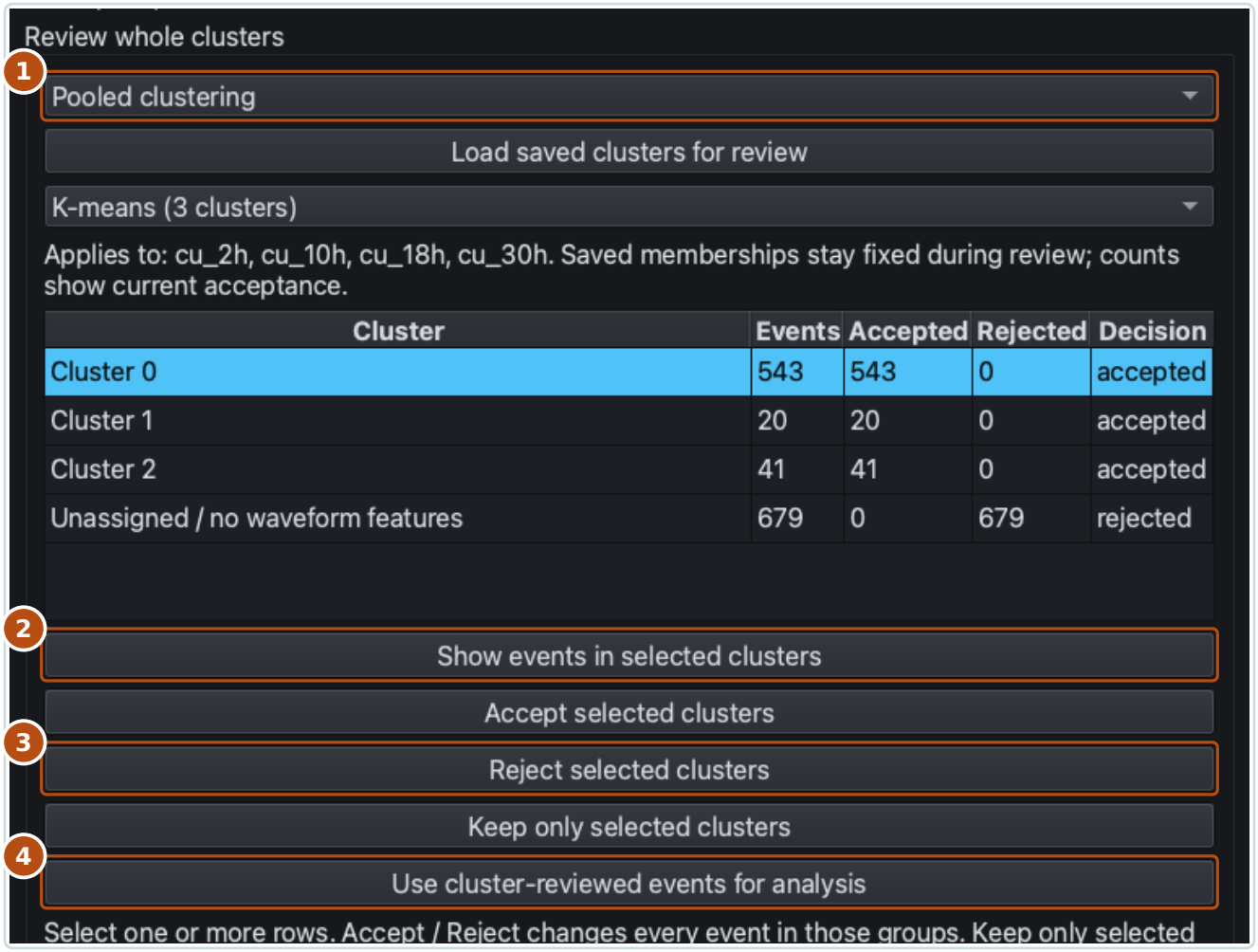
For reference reproduction, continue to comparisons without changing cluster acceptance. To curate groups, use the next page. K-means groups are descriptive; t-SNE is a visual display, not a quantitative separation test.

You should see

Unchanged reference: 604 primary feature observations, four retained PCs, three K-means groups and all observations labelled HDBSCAN Noise (-1).

Accept or reject entire clusters

Optional curated branch · inspect, decide, confirm, recompute



Reference capture, 21 Sep 2026. Controls · scroll down

DO THIS IN NANOSCOPE

1. **Choose the grouping and its domain**
   Choose Pooled clustering and K-means (3 clusters) or HDBSCAN. For older results click Load saved clusters for review. Read Applies to:.

2. **Inspect the group's events**
   Select cluster rows and click Show events in selected clusters. Inspect their waveforms.

3. **Apply your group decision**
   Accept selected clusters or Reject selected clusters changes those groups. Keep only selected clusters also rejects all other events in the displayed domain, including unassigned events.

4. **Confirm and rebuild the final results**
   Click Use cluster-reviewed events for analysis. Confirm other pending recordings, repeat feature extraction and final grouping, then recompute comparisons and the mixture.

Keep in mind

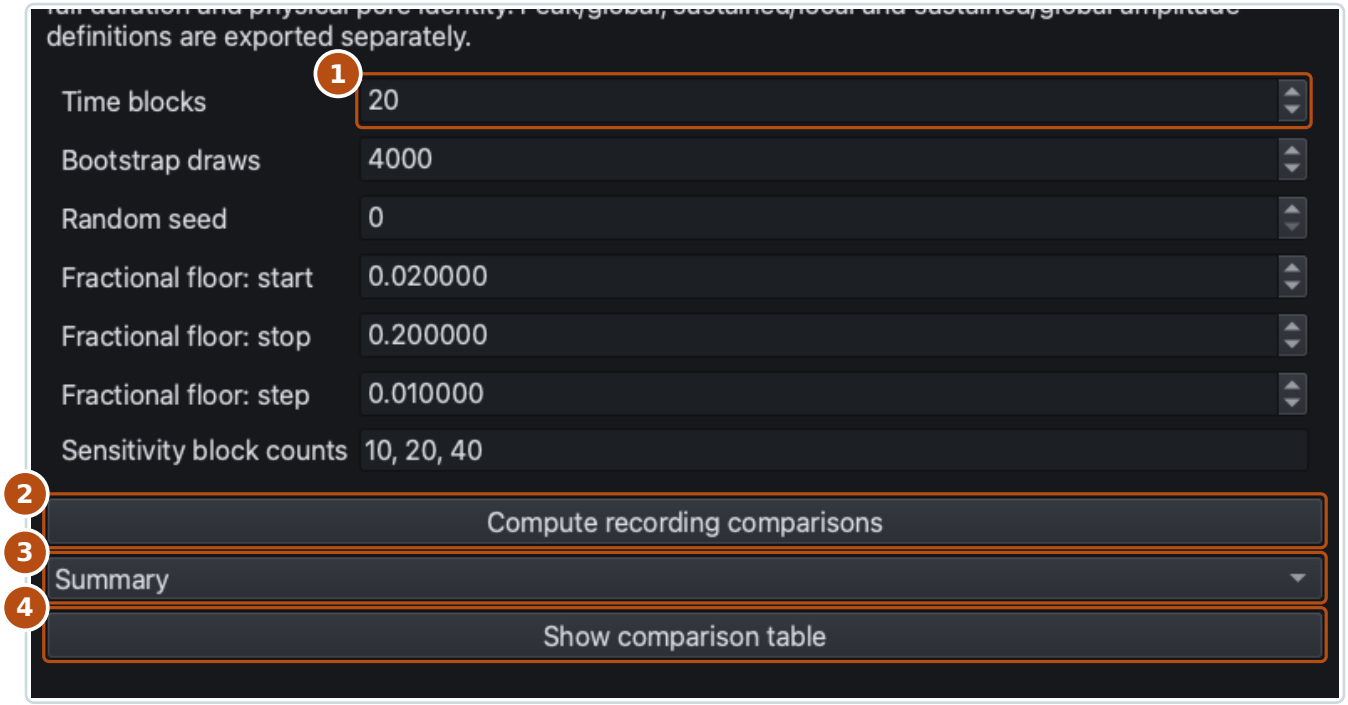
Reference HDBSCAN labels all 604 primary events Noise (-1); rejecting Noise removes them all. Unassigned is different. Appendix H explains snapshot persistence and mixture-specific curation.

You should see

Final analyses use your selected events. Record your reasons; counts may differ.

Compute the recording comparisons

4 Compare · use the final confirmed selection



Reference capture, 21 Sep 2026. Controls · top of panel

DO THIS IN NANOSCOPE

1. **Enter comparison settings**
   Open 4 Compare. Set Time blocks 20, Bootstrap draws 4000 and Random seed 0. Retain the remaining values in Recipe E.

2. **Run the comparisons**
   Click Compute recording comparisons and wait.

3. **View the summary**
   Select Summary and click Show comparison table. Check the four primary rows against Checkpoint 2 if reproducing the reference.

4. **Inspect supporting tables**
   Choose Accepted events, signal_types, nominal_pairs, drift and monotonicity in turn.

Keep in mind

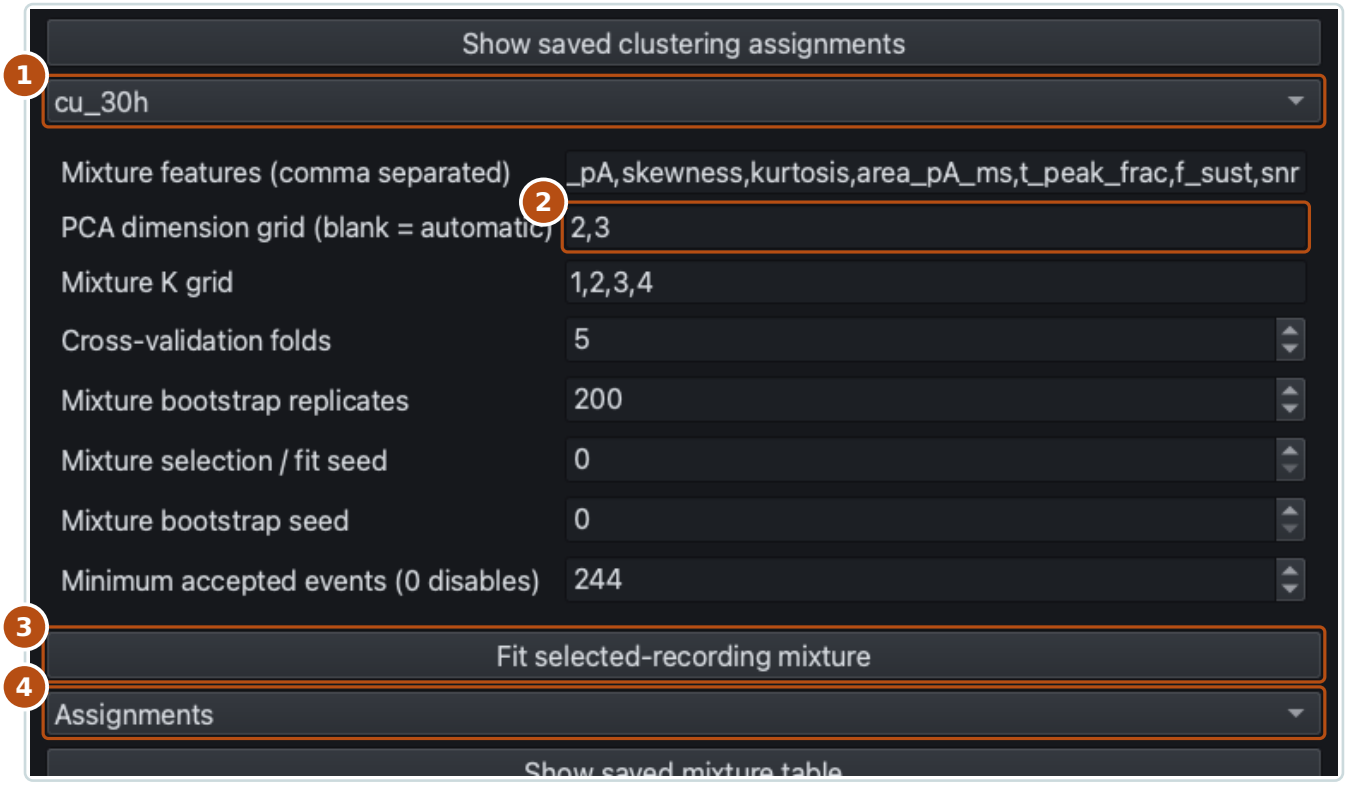
The fractional-floor and sensitivity fields serve the Zn model route and impose no extra Cu acceptance gate. Cu’s spike/plateau rule does not have the single sustained-depth gate-transfer tables. Rates use retained exposure.

You should see

The unchanged reference summary has 34, 95, 58 and 417 accepted events at 2, 10, 18 and 30 h. The supplementary excerpt is not a fifth pooled timepoint.

Fit the 30-hour population appendix

5 Features / populations · after final feature extraction



Reference capture, 21 Sep 2026. Controls · scroll down

DO THIS IN NANOSCOPE

1. **Choose the target**
   In the mixture target selector choose cu_30h.

2. **Enter the mixture recipe**
   Paste the same 12-feature list; set PCA grid 2,3, K grid 1,2,3,4, folds 5, bootstraps 200, both seeds 0 and minimum accepted events 244.

3. **Fit the mixture**
   Click Fit selected-recording mixture and wait for completion.

4. **Read results and warnings**
   Inspect all six saved-mixture tables and Methods / history. Compare the fit with Checkpoint 3 only if the reference selection is unchanged.

Keep in mind

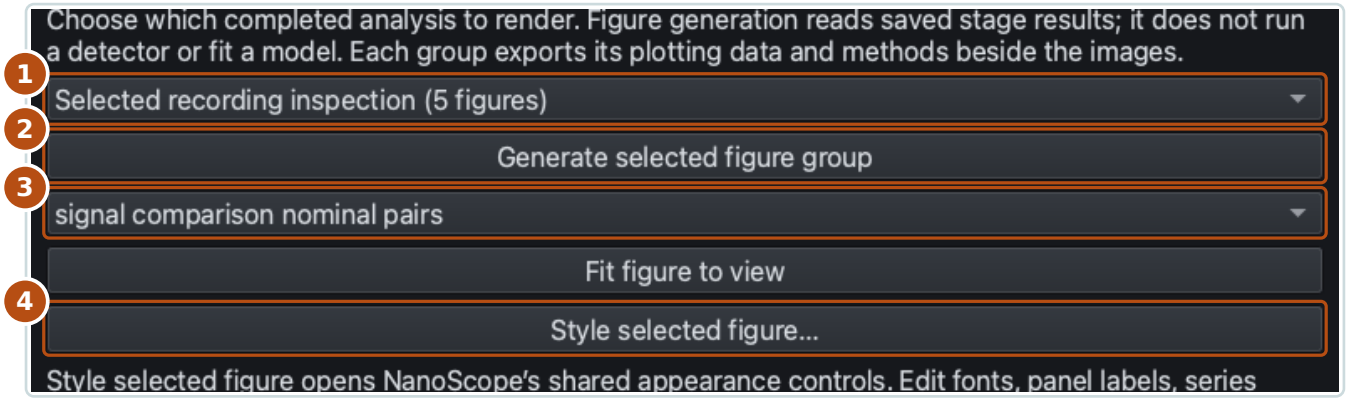
Both selected dimensions reach the tested grid boundary. Weak separation, temporal instability and bootstrap label switching limit interpretation. These are exploratory components, not four established molecular species. If curation leaves fewer than 244 events, do not silently lower the floor; a smaller justified specification is a new analysis.

You should see

Unchanged reference: 417 events, three PCs, four components; hard assignments 236 / 110 / 36 / 35.

Generate the complete 35-figure set

6 Figures · finish all scientific operations first



Reference capture, 21 Sep 2026. Controls · top of panel

DO THIS IN NANOSCOPE

1. **Generate each recording's inspection group**
   Choose Selected recording inspection (5 figures). Select each of the five recordings and click Generate selected figure group: 25 images total.

2. **Generate the three final analysis groups**
   Repeat Generate selected figure group for Recording comparisons, Pooled exploratory clustering and Selected-recording mixture: 5 + 2 + 3 images.

3. **Inspect each saved figure**
   Choose a plot in the saved-figure selector; it opens in Saved figure. Appendix G lists all 35 outputs.

4. **Start a styling session**
   Select the figure to edit and click Style selected figure....

Keep in mind

Event-example and null-control figures deliberately include candidate classes, including rejected shallow_long examples. Accepted distributions, rates, features and fits use your selection. After changed curation, regenerate current groups; older runs folders may contain obsolete images.

You should see

The current figure list contains 35 saved plots from the final results. No detector or fit is rerun by figure generation.

Choose the plot’s fonts and appearance

Figure editor · preview first, then save to the study



Reference capture, 21 Sep 2026. NanoScope reference controls and saved results.

DO THIS IN NANOSCOPE

1. **Choose a starting appearance**
   From 6 Figures, open Style selected figure.... In the editor, choose Publication · light or another appearance preset.

2. **Set the font**
   Choose Font and Background. Use a readable font for the final publication size.

3. **Set the text sizes**
   Adjust Title (pt), Axis labels (pt), Ticks (pt) and legend size.

4. **Set lines, markers and the grid**
   Adjust Grid, Line width, Marker size and legend visibility. Continue down the left panel for labels, colors and saving.

Keep in mind

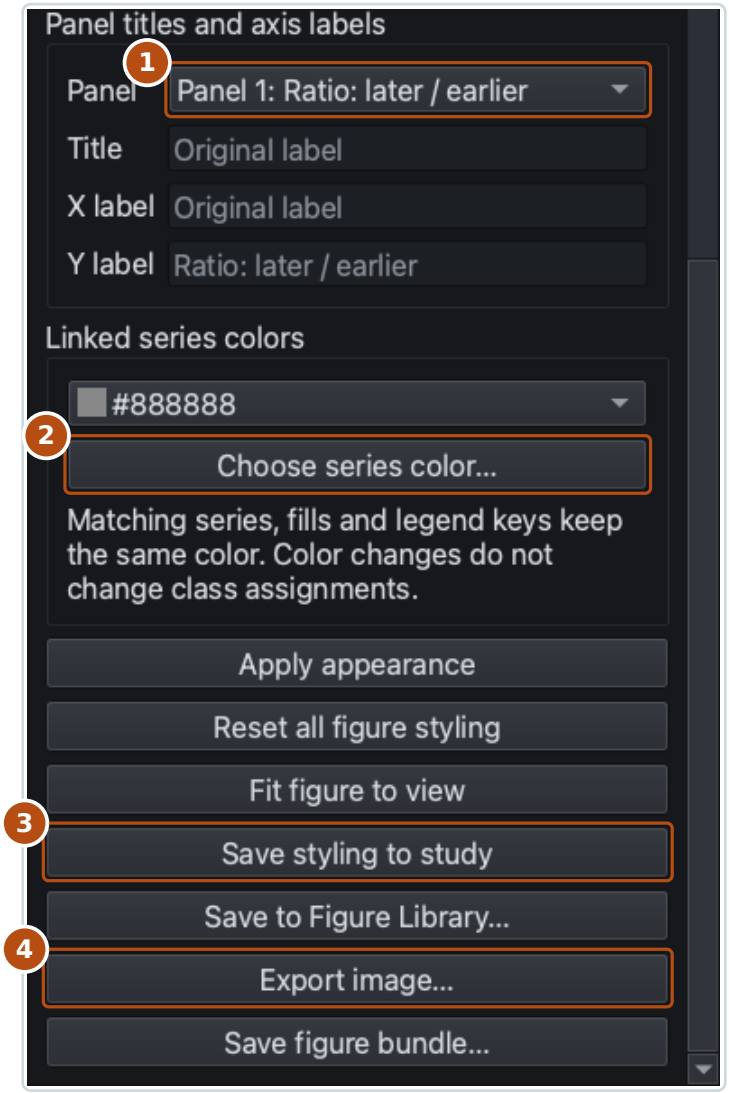
Appearance choices do not change numerical values, event selections or fitted populations. The editor does not expose arbitrary scientific axis ranges, axis scales or plot-type changes.

You should see

Your appearance choices are entered. Preview and save them on the next page.

Save labels, colors and final figure files

Figure editor · individual exports or frozen library entries



Reference capture, 21 Sep 2026. NanoScope reference controls and saved results.

DO THIS IN NANOSCOPE

1. **Edit the panel labels**
   Scroll down. In Panel titles and axis labels choose a panel and edit its title, X label or Y label. Preserve units and the time convention.

2. **Choose the linked colors**
   Select a series in Linked series colors and click Choose series color…. Its marks, fills and legend keys change together.

3. **Preview and save the style**
   Click Apply appearance and inspect the result. Click Save styling to study and wait for confirmation.

4. **Export the figure**
   Click Export image… for PNG, SVG or PDF; choose size and destination. Save figure bundle… also preserves numerical and provenance resources.

Keep in mind

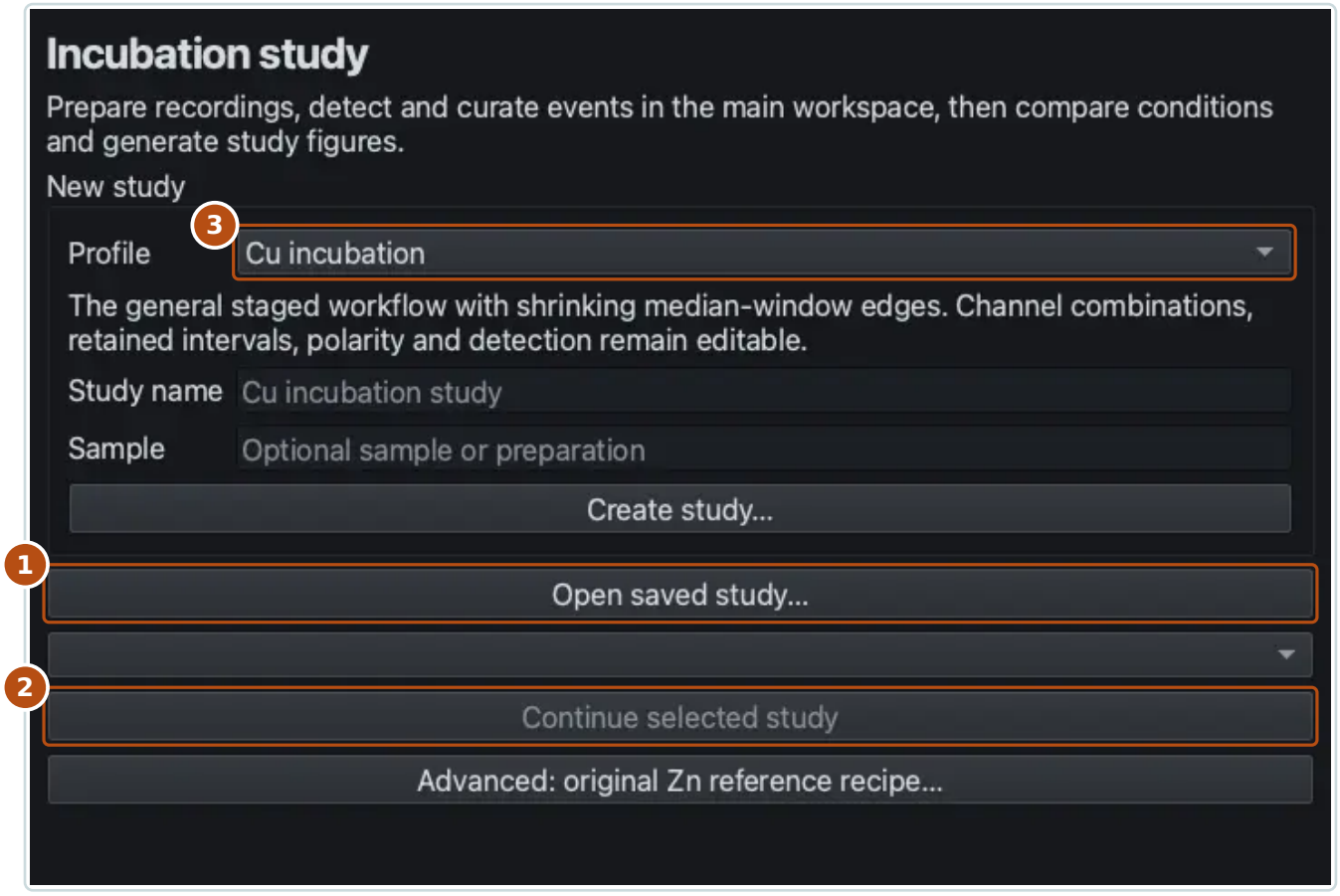
Apply alone only previews. Save to Figure Library… supports optional composition (Appendix I); save the main project separately. Recheck cluster colors after refitting.

You should see

The committed style survives reopening and regeneration; chosen exports are saved.

Save and reopen the complete study

Unified incubation workflow | NanoScope 40.15.59



Current unified study launcher · cu

DO THIS IN NANOSCOPE

1 Reopen the saved folder
In the study, finish any operation and click Save workspace. Close it, then open Analysis > Incubation study... > Open saved study.... Select the whole workspace folder.

2 Continue an open study
If the study is already open, select it in Open study workspaces and click Continue selected study.

3 Keep the saved method
The Profile control is for new studies. Opening a saved study selects the proper scientific method without applying a different profile.

Keep in mind

Keep workspace.json, arrays, runs and supporting files together. Preserve original ABFs and models for reruns. Use one editing session per folder. Save the main project separately for main curation and Figure Library entries.

You should see

Check restored event decisions, committed settings, comparison tables, fits and styled figures without rerunning completed stages.

A / Recipe A · files and reference locations

All reference inputs are 50,000 Hz. Desktop means /Users/abrahamcampos/Desktop on the reference computer. Select identical copies on another computer. The spelling adjuested is part of the real primary 30-hour filename.

The completed reference folder is /Users/abrahamcampos/Documents/Codex/2026-09-12/le/outputs/cu-study/manual-study. Analysis > Incubation study... > Open saved study... restores it for inspection. A fresh reproduction begins in a new empty folder. Never use the reference as scratch space.

Exact source identities

Tag	Location relative to Desktop	SHA-256
cu_2h	Cu2Data/100 pM Cu_2h_16 nm_50 PEG_0011_-200 mV_Conductive_Good_10-15 min.abf	3e413374ccb597d9c8316cb8b85ec30ec79c0d43c1e3aade6fed9c7a4bb10ad7
cu_10h	Cu2Data/100 pM_10 h_Cu_16 nm_50k PEG_0004_-200 mV_Conductive_good.abf	3ea561ac57129bb108aa9ede1840bca69e55280e82c66142ac76410296e4a7c9
cu_18h	Cu2Data/100 pM Cu_18 h_18 nm_50 PEG_0002_-200 mV_Conductive_Very Good_10-14.5 min.abf	f171c9be25925a293c8dfdf66c2a5aea46c2a41872c1035c30e6acda9d285d3c
cu_30h	Cu_30h_100 pM_18nm_50 PEG_0001_Process_-200 mV_Conductive_Very Good_Baseline adjuested.abf	f31745f7367165213ba8562acb073098084c87c2d59c66576199da8ccc3aa8f1
cu_30h_excerpt	Cu2Data/Cu_30h_100 pM_18nm_50 PEG_0001_Process_-200 mV_Conductive_Very Good_10-15 min.abf	7fe9374bd72ff424c1da3fe96187860c0c63981980b4938eba9d8e30a864497a

B / Recipe B · complete source settings

Apply recording details for each recording before leaving it. Keep intervals include the start and exclude the stop; exclusions remove both endpoints. All interval values are file-relative sample times, not the acquisition-header clock.

The separate physical pore IDs prevent matching nominal diameters being mistaken for the same physical pore.

Common to all five recordings

Control	Value
Concentration (pM)	100
Voltage (mV)	-200
Current channel (1-based)	1
Event orientation	Downward in source
ABF numeric scaling	Notebook float32 convention
Median window edges	Shrinking (Cu convention)

Per-recording settings

Control	cu_2h	cu_10h	cu_18h	cu_30h	cu_30h_excerpt
Incubation (h)	2	10	18	30	30
Physical pore ID	cu_2h	cu_10h	cu_18h	cu_30h	cu_30h_excerpt
Pore diameter (nm)	16	16	18	18	18
Keep from source time (s)	0	0	0	300	0
Keep until source time (s; 0 = end)	0	0	0	900	0
Exclude source intervals	blank	0:20;370:380	blank	blank	blank
Add current channel (1-based; 0 disables)	0	0	0	2	0
Additional channel values are pA (override header)	off	off	off	on	off
Supplementary recording (exclude from pooled series)	off	off	off	off	on

C / Recipe C · preparation

Enter for each recording

Prepare control	Value
Apply sequential notch	off
Notch frequencies (Hz, comma separated)	blank
Notch Q	30; unused while notch is off
Round notch output to float32	off
Baseline method	Event-masked median
Mask: rough window (s)	0.5
Mask: baseline window (s)	0.5
Median-grid decimation (samples)	10
Mask threshold (MAD sigma)	3
Mask dilation (ms)	5
Robust median / SavGol window (s)	0.5; unused for Event-masked median
SavGol polynomial order	3; unused for Event-masked median
Despike kernel (samples; 0 disables)	0

D / Recipe D · Cu signal detector and null controls

Candidate depth is measured on the unsmoothed prepared current. The entry threshold is the larger of 4 × MAD sigma and 8 pA. A short spike passes. A region at least 1 ms long needs sustained depth of at least 3 MAD sigma; otherwise it is shallow_long and initially rejected.

passes_gate retains the automatic decision; accepted is the later-analysis selection; event_review records explicit review actions. Even an accept/reject click that leaves the value unchanged records individual review evidence.

Schmitt signal controls

Control	Value
Signal mean window (samples)	5 (0.1 ms at 50 kHz)
Signal enter (MAD sigma)	4
Minimum enter threshold (pA)	8
Signal exit (MAD sigma)	1
Signal merge gap (ms)	5
Signal minimum duration (ms)	0.2 (10 samples)
Plateau minimum duration (ms)	1
Plateau sustained depth (MAD sigma)	3
Raw I0 rolling window (s)	5
Raw I0 grid (samples)	50

Null controls

Control	Value
Global-phase null seeds	5
Block-phase null seeds	3
Null block duration (s)	10

E / Recipe E · final recording comparisons

The first three controls set the Cu comparisons. Keep the remaining visible fields at the values below for the saved recipe; the Cu Schmitt route does not use the Zn floor/sensitivity fields to impose another acceptance gate.

Comparison settings

Control	Value
Time blocks	20
Bootstrap draws	4000
Random seed	0
Fractional floor: start	0.02
Fractional floor: stop	0.2
Fractional floor: step	0.01
Sensitivity block counts	10,20,40

F / Recipe F · pooled features and 30-hour mixture

Paste the following exact comma-separated list into both feature fields. Additional columns such as entropy may exist in the table but are not part of this reproduction:

log_dwell,depth_max_pA,depth_sust_pA,mean_pA,std_pA,min_dev_pA,skewness,kurtosis,area_pA_ms,t_peak_frac,f_sust,snr

Feature-stage f_sust uses whole-recording raw I0. The detector/comparison f_sust uses local raw I0. An identical column name across stages does not mean the measurement reference is identical.

The 244-event floor follows five training events per free parameter for the largest full-covariance model (3 PCs, 4 components) under fivefold cross-validation. A smaller curated set needs a newly justified specification. The 200 bootstrap fits estimate fitting variability, not how often events were reviewed.

The manual bundle includes Cu_features.txt. Copy its entire line into both feature-list fields to avoid retyping names.

Pooled clustering

Control	Value
Pooled clustering features (comma separated)	Exact list above
Clustering seed	0
K-means components	3
PCA retained variance target	0.95
Maximum retained PCs	10
HDBSCAN minimum samples	10
Compute descriptive t-SNE	on

Selected-recording mixture

Control	Value
Mixture target	cu_30h
Mixture features (comma separated)	Same exact list
PCA dimension grid (blank = automatic)	2,3
Mixture K grid	1,2,3,4
Cross-validation folds	5
Mixture bootstrap replicates	200
Mixture selection / fit seed	0
Mixture bootstrap seed	0
Minimum accepted events (0 disables)	244

checkpoints / Reference checkpoints · unchanged automatic selection

These counts apply before changed acceptance, boundaries or settings. A curated study is allowed to differ. Matching counts alone cannot establish waveform or boundary agreement.

Checkpoint 1 · retained exposure and detector output

Tag	Samples	Exposure (s)	Candidates	Accepted	Rejected
cu_2h	15,008,304	300.16608	60	34	26
cu_10h	18,711,198	374.22396	111	95	16
cu_18h	13,868,658	277.37316	161	58	103
cu_30h	29,992,491	599.84982	951	417	534
cu_30h_excerpt	15,076,224	301.52448	729	274	455

Checkpoint 2 · primary comparison

Tag	Accepted	Rate (events/s)	Median dwell (ms)	Median local fractional change (%)
cu_2h	34	0.113271	0.53	4.0418
cu_10h	95	0.253859	0.74	1.9286
cu_18h	58	0.209105	0.60	4.8809
cu_30h	417	0.695174	0.50	8.1051

Checkpoint 3 · features and mixture

Result	Reference
Primary accepted / pooled feature observations	604
30-hour mixture observations	417
Pooled PCs for 95% target	4
Pooled HDBSCAN density clusters	None; all 604 Noise (-1)
Selected mixture	3 PCs, 4 components
Hard assignments	236 / 110 / 36 / 35
Mean silhouette	Approximately 0.325
Fit cautions	Grid boundaries; weak separation; temporal instability; bootstrap label switching

G / Appendix G · all 35 saved figures

Numbers below are report order. NanoScope identifies figures by saved names; its selector shows stems with spaces. Generate every required group after the final analyses. Each PNG is accompanied by plotting-data and methods resources under runs.

The five recording figures are overview, distributions, candidate examples, null controls and time dependence. Candidate examples and null controls are diagnostic and can include rejected candidates. Rates, accepted distributions and populations use accepted events.

The assigned 30-hour trace (figure 35) marks actual minima inside assigned events in its upper view and their onsets below. It uses retained time: add 300 s for the source ABF clock.

Figure inventory

Report figure	Group / recording	Saved filename
1	cu_2h	signal_cu_2h_trace.png
2	cu_2h	signal_cu_2h_distributions.png
3	cu_2h	signal_cu_2h_examples.png
4	cu_2h	signal_cu_2h_null_controls.png
5	cu_2h	signal_cu_2h_time_dependence.png
6	cu_10h	signal_cu_10h_trace.png
7	cu_10h	signal_cu_10h_distributions.png
8	cu_10h	signal_cu_10h_examples.png
9	cu_10h	signal_cu_10h_null_controls.png
10	cu_10h	signal_cu_10h_time_dependence.png
11	cu_18h	signal_cu_18h_trace.png
12	cu_18h	signal_cu_18h_distributions.png
13	cu_18h	signal_cu_18h_examples.png
14	cu_18h	signal_cu_18h_null_controls.png
15	cu_18h	signal_cu_18h_time_dependence.png
16	cu_30h	signal_cu_30h_trace.png

Report figure	Group / recording	Saved filename
17	cu_30h	signal_cu_30h_distributions.png
18	cu_30h	signal_cu_30h_examples.png
19	cu_30h	signal_cu_30h_null_controls.png
20	cu_30h	signal_cu_30h_time_dependence.png
21	cu_30h_excerpt	signal_cu_30h_excerpt_trace.png
22	cu_30h_excerpt	signal_cu_30h_excerpt_distributions.png
23	cu_30h_excerpt	signal_cu_30h_excerpt_examples.png
24	cu_30h_excerpt	signal_cu_30h_excerpt_null_controls.png
25	cu_30h_excerpt	signal_cu_30h_excerpt_time_dependence.png
26	Recording comparisons	signal_comparison_summary.png
27	Recording comparisons	signal_comparison_nominal_pairs.png
28	Recording comparisons	signal_comparison_distributions.png
29	Recording comparisons	signal_comparison_event_types.png
30	Recording comparisons	signal_comparison_excerpt_check.png
31	Pooled exploration	pooled_pca.png
32	Pooled exploration	pooled_tsne.png
33	cu_30h mixture	quantitative_appendix_selection_fractions_waveforms.png

Report figure	Group / recording	Saved filename
34	cu_30h mixture	population_examples_cu_30h.png
35	cu_30h mixture	population_onsets_cu_30h.png

H / Appendix H · interpretation and curation rules

• **Reference versus curated analysis:** Keep automatic decisions for reference reproduction. Inspect and confirm each recording even when retaining them. For a custom study, document every scientific selection rule. Individual review and whole-cluster review modify the same accepted field; the last action affecting an event determines its selection.

• **After a cluster decision:** Acceptance changes clear dependent scientific results but keep the previous membership snapshot so a group can be accepted again. Reconfirm, re-extract final features, rerun final grouping and mixture, recompute comparisons, then regenerate figures. New detections, remeasurement or native import retire old memberships. A new fit replaces that source’s snapshot; cluster numbers and colors may acquire different meanings.

• **Read the domain before Keep only:** Keep only selected clusters affects every event in the displayed recording domain, including unassigned events unless their bucket is selected. Other recordings remain unchanged. A selected-recording mixture snapshot for cu_30h affects only cu_30h. The 244-event floor remains in force after curation.

• **Match the event and its clock:** Manual preview, native review, rate time blocks and the 30-hour population-onset figure use retained acquisition time. Exported original_onset_s and original_end_s map back to the source file. Add 300 s for cu_30h; use the original-time columns for cu_10h because two removed intervals make one constant offset incorrect. Signal overview figures use original source time where labelled. Match event ID as well as the clock.

• **Limits on biological claims:** One recording per timepoint and different pore conductances limit interpretation of incubation trends. Primary 30-hour median raw I0 is about 77.19 pA; the supplementary excerpt is about 124.8 pA. K-means partitions do not establish three species, and the four-component mixture does not establish four molecular populations. Hard-assigned fractions, fitted mixture weights and bootstrap intervals are different quantities.

• **Current notebook versus older cache:** The original validation matched fresh NanoScope raw samples, prepared traces, inclusive event bounds, automatic classes, source timestamps and peak depths exactly against independent execution of the current notebook definitions. The old CSVData_mm3 cache has different prepared baselines and some different boundaries despite equal accepted counts. An old cached image alone does not disprove a fresh event table.

• **What this manual covers:** This guide reproduces the Cu incubation analysis and waveform-population appendix from raw ABFs, without a trained model. Teacher/student training and the separate IntraEventAnalysis encoder, regression and temporal-disruption experiments are different workflows.

• **Styling and caption identity:** Appearance controls change fonts, colors, lines, markers and labels, not arbitrary scientific axis ranges, axis scales or plot types. Zero/default appearance fields preserve original styling. Custom event-gallery captions follow the same event and trace; a changed or replaced event uses an automatic caption until restyled. Older captions without event identity must be re-entered; general styling remains available.

I / Appendix I · optional multi-panel composition

Save at least two figures from the study editor using Save to Figure Library…. In the main window choose File → Save Project… to preserve those entries. These steps use the main Figure Library, separate from the study workspace.

• **1. Open the composer:** Choose Analysis → Figure Library…, then Compose multi-panel figure….

• **2. Choose and arrange panels:** Check Use for 2–12 saved figures. Set labels, Columns, Rows / Cols spans and Panel spacing. Use Move up / Move down to order panels. Study figures can already contain several panels; keep their axes and legends unless the selection supports compatible sharing.

• **3. Inspect and save:** Click Update preview, inspect the layout, then Save to Figure Library. Save the main project again.

• **4. Export or revise:** Select the composition in the library and choose Export saved figure… or Save figure bundle…. To change it later, use Revise composition… and save a new revision.

• **5. Update after scientific changes:** Library entries are frozen. Save new source entries and revise the composition after changed curation, analyses or styling.

troubleshooting / Troubleshooting and final audit

Before completion, check identities and recipe values, review status and accepted counts, representative waveforms against source traces, final tables and fit diagnostics, all 35 figures, saved styles and successful reopening. Record any departure from the reference.

An older completed study can be viewed before its saved selection is confirmed. Opening it does not approve the events automatically. Before a new downstream analysis, inspect and confirm every included recording with Use reviewed events for analysis.

If this happens

Symptom	Action
The menu or a control is missing	Verify the full build ID in Help → About NanoScope…. Scroll the Analysis menu or the left control panel; widen the controls using the central divider.
Downstream analysis says review pending	Confirm every included primary recording with Use reviewed events for analysis, or confirm the displayed cluster domain. Check the supplementary recording separately.
Settings differ from completed results	A field edit or Save workspace does not rerun science. Restore completed settings returns to saved values; otherwise run the changed stage and its dependents. A draft on another recording can also block a stage.
Reference counts disagree	Check exact 30-hour file and channel reconstruction, 10-hour cuts, shrinking median edges, baseline settings, orientation and the Schmitt route. Inspect bounds as well as counts.
A rejected group seems missing	Use its original review source/method and check Applies to:. Acceptance retains memberships; detection, measurement or native import requires new grouping.
Native edits are absent	Use Save curation to study on the main study bar, then Return to study. Inspect and confirm the saved selection again.
Only some figures are listed	Finish science first and generate every figure group. Old files in runs may describe a prior selection.
Styling disappears	Apply appearance previews only; click Save styling to study. Save Project separately for library entries and compositions.
A newer workspace revision is reported	Use one editing session per folder; reopen to load the newest saved revision rather than overwriting it.
A saved workspace file is missing or changed	Restore the complete folder from an intact copy, including workspace.json, arrays, runs and supporting files.

SHARED / Shared workflow for Zn, Cu, acid and future studies

All studies open through Analysis > Incubation study.... General, Zn and Cu use the editable staged workspace. Acid retains its calibrated CUSUM method and frozen-family definitions. Changing the entry point does not make their scientific detectors interchangeable.

For a future experiment choose General incubation, name the sample and conditions, and enter the actual time, pore identity, concentration, voltage, channel and retained intervals. Choose and document preparation, detection and measurement settings appropriate to that study. A model is optional; selecting a profile does not validate a new experiment.

The main event table, Gallery and Inspector are shared. Save curation to study commits a snapshot. Return, confirm the selection and recompute dependent analyses. Keep automatic and human decisions distinguishable, and do not treat clusters as molecular identities.

Advanced: original Zn reference recipe... is the fixed automated reference route. Its automated run uses the fixed recipe. For curated analysis, save events to the staged study, confirm the selection and run that study's downstream stages.